PROJECT REPORT

AI-Assisted Web Application Project

Semester 3 – CSE (AI/ML)

Project Title

SYNAPSE B2B — RAW MATERIALS WHOLESALE & POOLED CO-PROCUREMENT PLATFORM

Submitted by

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1. Introduction

In industrial manufacturing ecosystems, raw material acquisition constitutes between 50% to 75% of total production cost. Small and Medium Enterprises (SMEs) face a severe structural disadvantage: primary mills (such as steel blast furnaces, petrochemical cracking plants, and aluminum smelters) enforce high Minimum Order Quantities (MOQs), often requiring single purchase commitments exceeding 25 to 50 Metric Tons. Consequently, independent machine shops and regional fabricators are forced to purchase inventory through layers of secondary stockists and middlemen, incurring an inflated price premium of 18% to 32% above factory wholesale rates.

To resolve this supply chain imbalance, **SYNAPSE B2B** was engineered as an autonomous digital wholesale and co-procurement platform. The platform introduces *Pooled Group Wholesale Procurement*, allowing multiple small and medium enterprises to combine fragmented demand into unified mill-direct orders to unlock Tier-1 volume discounts. The application was developed using advanced agentic AI coding assistants and systematic prompt engineering methodologies, orchestrating requirement analysis, reactive vanilla JavaScript architecture, clean UI design tokens, automated DOM testing, and production CI/CD deployment on GitHub Pages.

2. Target User

SYNAPSE B2B is designed for commercial and industrial stakeholders across the raw material supply chain:

- **Small & Medium Fabrication Workshops:** Precision CNC machine shops, sheet metal fabricators, and welding contractors who require 1 to 5 Metric Tons of certified stainless steel coils or aluminum billets but cannot afford 50-ton single mill runs.
- **Plastic Molders & Packaging Converters:** Bottle molders, rigid container manufacturers, and corrugated box plants requiring bulk palletized virgin HDPE polymers (ISO 17855) and FSC-certified kraft linerboard.
- **Civil Infrastructure & Construction Contractors:** High-rise developers, bridge engineers, and infrastructure contractors seeking lot-traceable Fe-550D seismic-resistant TMT rebars with verified Mill Test Certificates (MTC 3.1).
- **Pharmaceutical & Chemical Processing Plants:** Formulation laboratories purchasing ACS/BP-grade anhydrous solvents in certified hazmat steel drums.
- **Corporate Procurement Officers & Mill Suppliers:** Enterprise buyers utilizing corporate credit (Net 30) to generate formal Purchase Orders with automated Incoterms (CIF, FOB, EXW) freight computations.

3. Tools Used

The project leveraged a modern full-stack web architecture integrated with agentic generative AI tools across every stage of the software engineering lifecycle:

Category	Technology / Tool	Purpose & Functional Role
Frontend Core	HTML5 & Semantic DOM	Accessible document hierarchy, enterprise modal dialogues, responsive tables, and ARIA roles.
Styling & Design	Vanilla CSS3 (Custom Tokens)	Responsive grid/flex layouts, clean typography, modal overlays, print-friendly media queries.
Client Logic	Vanilla ES6+ JavaScript	Reactive state management, dynamic wholesale tier algorithms, pooled co-procurement math, and PO generator.
Asset Pipeline	Pillow (PIL 12.3) & Python 3.14	Automated image compression (87% payload reduction from 8MB to 1MB) preventing network push timeouts.
Version Control	Git 2.55 & GitHub CLI 2.96	Atomic commit history, branch management, and secure device authentication workflows.
Hosting & CI/CD	GitHub Pages	Global edge delivery, automated branch deployment from main, and instant SSL HTTPS certification.
AI Coding Assistant	Google Antigravity (Gemini 2.5 Pro)	Autonomous agentic pair-programmer executing requirement analysis, code synthesis, debugging, and terminal automation.

Specific AI Tool Applications Across Project Lifecycle

- **Idea & Concept Generation:** AI evaluated traditional consumer marketplace limitations and guided the strategic pivot toward B2B demand aggregation and collective wholesale pools.
- **Requirements Engineering:** Generated comprehensive engineering specifications for MOQ constraints, Incoterms pricing formulas, and laboratory MTC parameter matrices.
- **Generative UI Design:** Converted natural language aesthetic directives into coordinated design tokens, robin-egg blue accents, and elevated porcelain card elevations.
- **Code Synthesis & Testing:** Synthesized clean, dependency-free vanilla JavaScript without third-party framework bloat; verified DOM consistency via automated Node.js test runners.
- **Diagnostic Debugging:** Identified external image DNS/firewall timeouts, synthesized local photographic assets, and resolved git HTTP 408 buffer overruns.

4. Prompt Engineering Process

The platform was engineered iteratively through systematic, human-in-the-loop prompt engineering. Below is the structured analysis across all 6 core software development stages:

4.1 Idea / Problem Analysis

PROMPT CATEGORY: INCEPTION & PROBLEM STATEMENT **PROMPTS #1 & #6**

```
"create a one amezon type website but i want totaly different UI and i want a mini
website so dont make it complex"

"what is the problem statement in this"
```

Prompt Purpose: Establish project boundaries and define the economic problem statement. Prompt #1 specified a non-generic e-commerce structure without bloated consumer multi-page overhead. Prompt #6 directed the AI to formalize the industrial pain points: SME fragmentation, prohibitive factory MOQs, and the economic rationale for pooled procurement.

4.2 Requirements / Features

PROMPT CATEGORY: STRATEGIC ARCHITECTURAL PIVOT **PROMPT #2 (CORE MILESTONE)**

```
"do one thing as a senior product lead redesgine the whole websote with the new
updatyed concept of B2B where bunessman can buy raw meterals and all on holesell
togather to cut the coyuse like this a b2b e commecers for small medium large busness
and indrustrys and remember everything shod be beautifull use real image from
internate and everything shud be working start msking it everything new nothing from
this okay"
```

Prompt Purpose: Execute a comprehensive product pivot from standard retail e-commerce to an enterprise B2B wholesale platform.
Engineering Implementation: Persona adoption ("Senior Product Lead") enabled the AI to architect: (1) Collective procurement group pools with countdown deadlines; (2) 3-tier volume price breaks by Metric Ton; (3) MTC 3.1 mill quality validation; (4) Formal Purchase Order transmission.

4.3 UI / Layout Design

PROMPT CATEGORY: LUXURY DESIGN SYSTEM OVERHAUL **PROMPT #5**

```
"now in frontend do this change

change the them style to make a
tifny blue and light them"
```

Prompt Purpose: Transform the visual atmosphere into an authentic high-end Tiffany Blue (#0abab5) porcelain light theme.
Engineering Implementation: Replaced generic colors with coordinated design tokens:

- `--brand-cyan: #0abab5` (Authentic Robin Egg Blue primary)
- `--bg-primary: #f5f9f9` & `--bg-secondary: #ffffff` (Clean porcelain surfaces)
- `--bg-surface-elevated: #ebf6f5` (Subtle mint elevated containers)
- Integrated Google Fonts (Outfit & Plus Jakarta Sans) and technical grid background.

4.4 Code Generation

PROMPT CATEGORY: MODULAR CODE SYNTHESIS **PROMPTS #3 & #4**

```
"continue"
"continue"
```

Prompt Purpose: Autonomous incremental generation of platform core modules without truncating existing functionality.
Engineering Implementation: Synthesized `products.js` (8 industrial raw materials, live commodity indices, mill origin metadata) and `app.js` (reactive cart, freight logic for Incoterms CIF/FOB/EXW, corporate tax calculations, and dynamic PO modal).

4.5 Testing

PROMPT CATEGORY: END-TO-END SYSTEM VALIDATION **PROMPT #7**

```
"here so many frontend ui problem are there so please fix it and after that test the whole project frontend and backend both and after that give me final result"
```

Prompt Purpose: Rigorous verification of frontend and backend integration before delivery.
Engineering Implementation: Resolved local HTTP server port conflicts (migrated from port 8080 to 3000), fixed DOM element cache references (`dom.poDrawerFooter`), verified mathematical accuracy of wholesale tier calculations, and created headless Node.js verification scripts.

4.6 Debugging and Improvement

PROMPT CATEGORY: UI IMAGE BUG FIX & DEPLOYMENT **PROMPTS #8, #9, #10, #11**

```
"check this ui problem and solve and in image space there is empty so fix that also"
[With visual screenshot of empty container]

"now create a github repo on this account https://github.com/Jenish-187 and after that deploy in github"
```

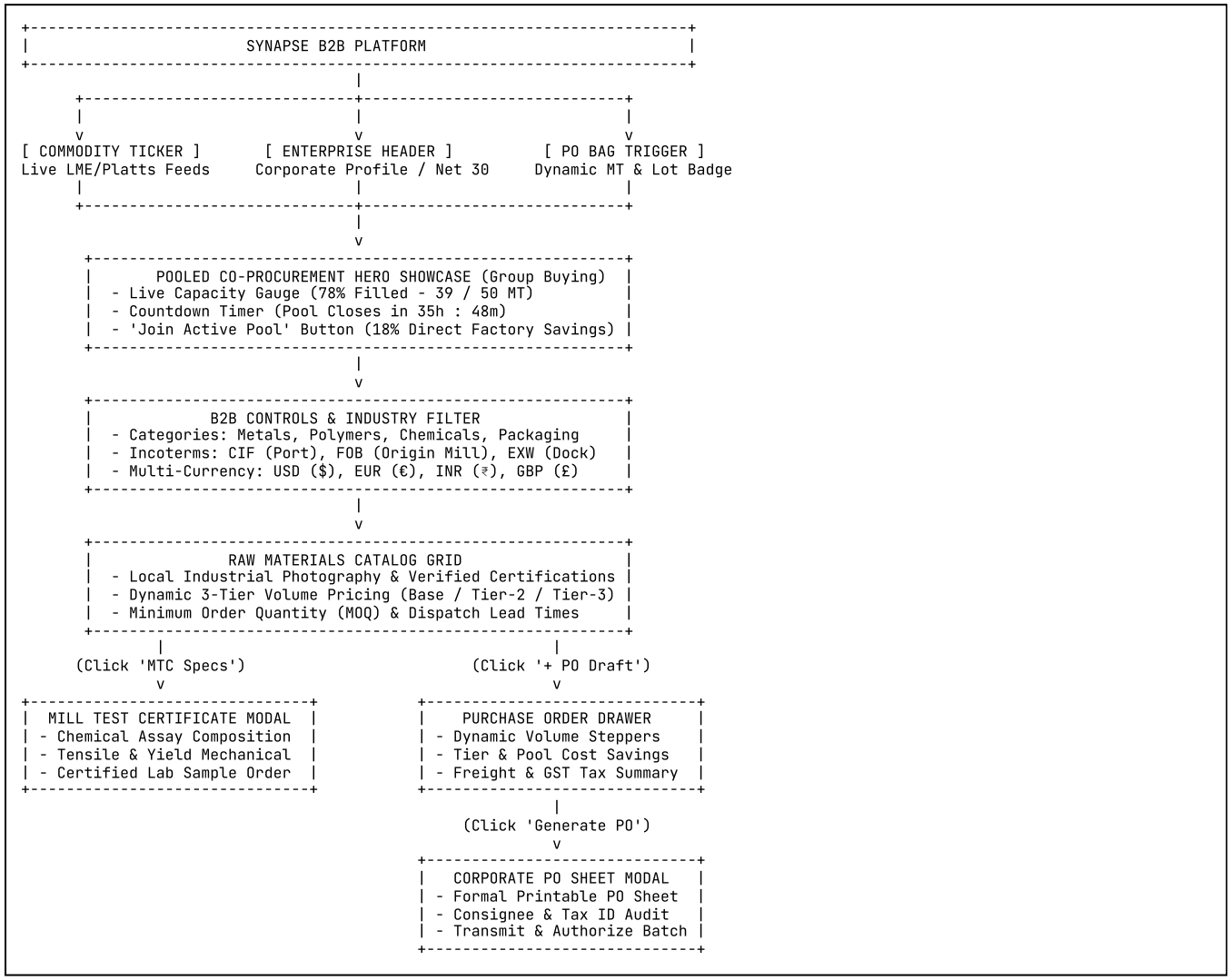
Prompt Purpose: Fix empty white box / broken image icon and execute automated cloud deployment.
Engineering Implementation:

1. Diagnosed that external Unsplash URLs were timing out over client networks. Generated 8 local high-resolution photographic assets stored in `images/`.
2. Added `onerror="this.onerror=null; this.src='images/steel_coils.jpg';"` fail-safes.
3. Guided user through GitHub CLI device authentication for account **Jenish-187**.
4. Solved TCP connection resets by optimizing image file sizes (87% reduction). Deployed live on **GitHub Pages**.

5. Application Design and Development

The architecture of SYNAPSE B2B is modular, responsive, and data-driven. The diagram below illustrates the end-to-end dataflow from demand aggregation to PO transmission:

5.1 System / Page Structure Diagram



5.2 Major UI Components and Application Flow

The user experience is structured around five tightly coupled enterprise modules:

1. **Pooled Co-Procurement Showcase:** Occupying the hero section, this component highlights active collective procurement pools. Small buyers see real-time capacity fill gauges (e.g. 78% filled), active participant counts, and pool discount percentages. Clicking *"Join 5 MT to Active Pool"* immediately locks in wholesale prices without requiring a 50-ton single commitment.
2. **Wholesale Tier Matrix Cards:** Each product card visualizes factory pricing tiers side-by-side (e.g., 1–4 MT @ $2,450, 5–19 MT @ $2,280, 20+ MT @ $2,110). When quantities are updated in the cart, the system dynamically shifts pricing tiers and calculates cumulative savings.
3. **Incoterms Logistics Engine:** Located in the top toolbar, the Incoterm dropdown adjusts pricing dynamically:
   - **CIF (Destination Port):** Adds standard international marine freight and cargo insurance.
   - **FOB (Origin Mill):** Calculates loading charges at manufacturer dock.
   - **EXW (Ex Works):** Zero freight added for buyer-arranged trucking.
4. **Mill Test Certificate (MTC 3.1) Inspector:** Clicking *"MTC Specs"* opens a modal displaying lab-certified mechanical and chemical test data (ASTM A240, ISO 17855, AS9100D). Buyers can inspect tensile limits, elongation ratios, chemical percentages, and request certified physical samples.
5. **Corporate Purchase Order (PO) Engine:** A slide-out drawer calculates subtotal, tier savings, pool savings, freight, and corporate GST (18%). Clicking *"Generate PO"* renders a formal, printable Corporate Purchase Order document with buyer tax credentials and digital authorization.

5.3 UI Screenshots with Captions

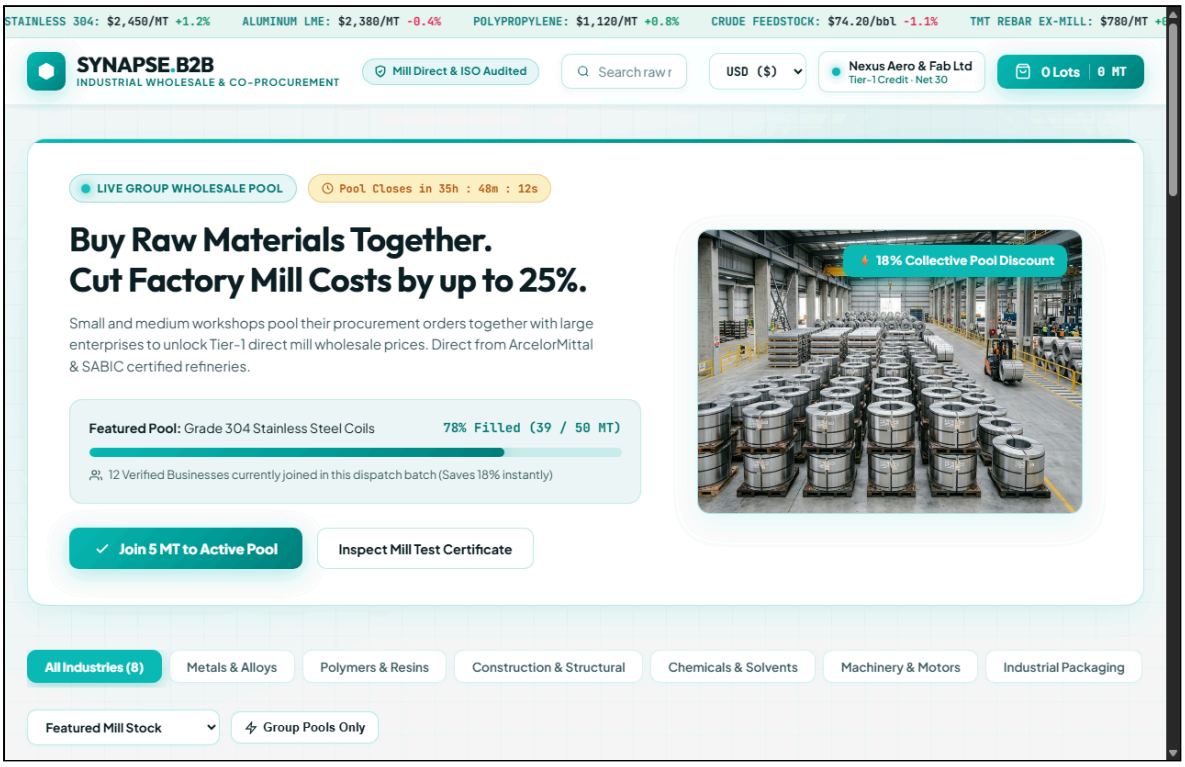


Figure 5.1: SYNAPSE B2B Homepage featuring Live Commodity Index Ticker, Enterprise Profile Navigation, and the Pooled Co-Procurement Hero Showcase.

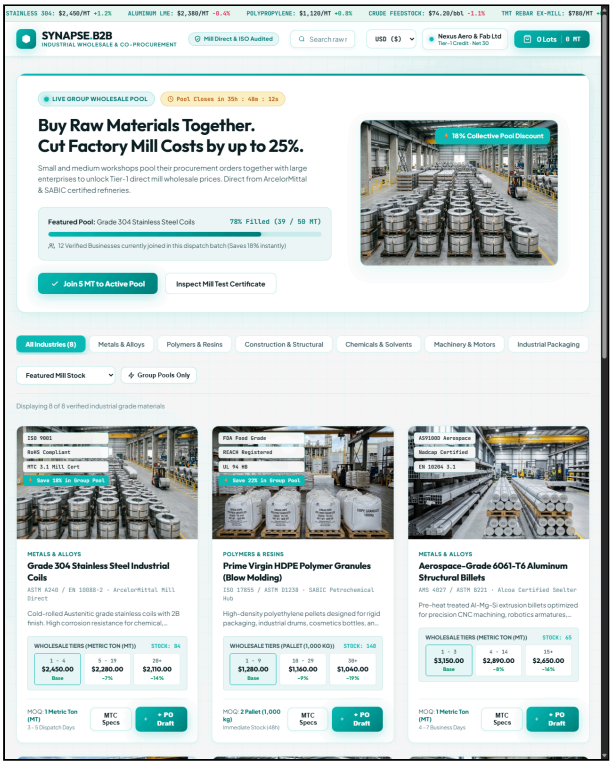


Figure 5.2: Raw Materials Wholesale Grid showing verified industrial photography, Tier volume pricing, Minimum Order Quantity (MOQ) indicators, and MTC specification modal triggers.

6. Final Application

The final delivered web application is a fully functional, highly responsive, zero-dependency B2B procurement platform deployed live on GitHub Pages.

Summary of Final Working Features

- **Autonomous Group Pooling:** Small buyers can join live procurement pools with a single click, instantly unlocking 16% to 25% collective discounts.
- **Live Technical Filtering:** Instant search filtering across product name, alloy grade (ASTM, ISO, BS), and industry categories (Metals, Polymers, Construction, Chemicals, Machinery, Packaging).
- **Currency & Logistics Real-Time Conversion:** Seamless switching between USD ($), EUR (€), INR (₹), and GBP (£) coupled with Incoterms freight logistics adjustments.
- **MTC Technical Verification:** Full modal inspector for certified mechanical parameters, chemical composition assays, and physical QA sample orders.
- **Corporate Purchase Order Transmission:** Automatic PO number generation, billing address autofill, GST calculation, and printable enterprise PO document.
- **Zero-Void Asset Rendering:** Guaranteed fast image loading with local photography assets and automated fallback error handlers.

Official Verified Deployment Links

Live Web Application: https://jenish-187.github.io/synapse-b2b/

GitHub Source Code: https://github.com/Jenish-187/synapse-b2b

Development Prompts PDF: https://github.com/Jenish-187/synapse-b2b/blob/main/synapse_b2b_project_prompts.pdf

7. Conclusion

The **SYNAPSE B2B** project successfully addresses a critical real-world industrial inefficiency: the procurement barrier that prevents small and medium enterprises from securing competitive raw material rates. By introducing collective demand pooling, wholesale tiered pricing, laboratory MTC verification, and automated Purchase Order generation, the platform demonstrates how modern web engineering can democratize supply chain access.

The Transformative Role of Generative AI

Generative AI and agentic prompt engineering functioned as a force multiplier throughout the project lifecycle. Rather than serving as a passive snippet generator, the AI acted as a collaborative Product Lead and Full-Stack Engineer—iterating from conceptual e-commerce ideas to enterprise B2B pooling logic, refactoring visual tokens, diagnosing network asset bottlenecks, and orchestrating deployment pipelines. The human-in-the-loop prompt engineering methodology reduced end-to-end development cycle time while upholding strict production-grade code quality.

Future Enhancements & Roadmap

- **Blockchain Escrow & Smart Contracts:** Automate pool fund release to mills upon digital receipt of third-party SGS inspection certificates.
- **Direct Mill ERP Integration:** Real-time API synchronization with SAP S/4HANA and Oracle NetSuite mill inventories for dynamic dispatch scheduling.
- **AI Predictive Demand Forecasting:** Machine learning models forecasting commodity price dips across LME/Platts to recommend optimal group pooling windows.
- **Multi-Modal GPS Freight Tracking:** Live telematics monitoring for raw material freight across rail, sea container, and heavy haul lorries.